

Shortcut Solver — Exact Shortcut Selection on a Fixed Ring (MILP)

Real routed shortcut geometry: the stage that produces achievable objective values

Goal. For one given, fixed ring, choose the globally optimal set of shortcuts so that the worst communicating pair has the shortest possible path. Unlike the proxy master, the Shortcut Solver uses real routed shortcut geometry, so its objective $W_{\text{true}}(R)$ is a genuinely achievable value on that ring.

1 Setup

The ring R is an input and is never modified: its tour order and the L-shape orientation of every ring edge are fixed. Node coordinates (x_i, y_i) , Manhattan distance $\delta(i, j) = |x_i - x_j| + |y_i - y_j|$, and the demand set Q with $s(q), t(q)$ are as in the master. Write $\text{ringDist}_R(i, j)$ for the shorter of the two arc lengths between i and j along the fixed ring. The method has two strictly separated phases.

Intuition

The routing grid is built from the ring alone — ring edges are the obstacles. It does not depend on which shortcuts are eventually chosen. Therefore every candidate's geometry can be computed once, up front, and the optimisation that follows is purely combinatorial: pick a subset of pre-measured bridges. Selecting one shortcut never changes another's length, only whether the two may coexist.

2 Phase 1 — Geometric precomputation (no optimisation)

Executed once for the given ring.

(P1) Candidate set. Unordered node pairs $\{i, j\}$, excluding

- pairs that are ring neighbours (a shortcut cannot beat the single ring edge already there);
- dominated pairs with $\delta(i, j) \geq \text{ringDist}_R(i, j)$. Since the real routed length satisfies $L \geq \delta$, such a shortcut could never appear on a shortest path. Dropping them is safe and loses no optimal solution.

(P2) Grid. One rectilinear routing grid is built from the ring only, and reused for all candidates. It is never rebuilt per candidate and never depends on a shortcut.

(P3) Routing. Each surviving candidate is routed in isolation: only the ring acts as an obstacle; other candidates are not treated as obstacles and no crossing rule is applied yet. Ring segments incident to the candidate's own endpoints are exempt, so a shortcut may leave and enter its own endpoints. For each candidate c this yields

- feasibility — infeasible candidates are discarded permanently;
- the routed path, its length L_c and its bend count β_c .

Let C be the set of feasible candidates. By construction $L_c \geq \delta(c)$.

(P4) Crossing matrix. For every pair of feasible candidates, test whether their stored paths intersect — a pure segment-intersection test on known geometry, no re-routing. Write $\mathcal{X}(c) \subseteq C$ for the candidates crossing c .

After Phase 1 no geometry is ever recomputed. The pair $(\{L_c\}, \mathcal{X})$ is the complete input to Phase 2.

3 Phase 2 — Variables

$z_c \in \{0, 1\}$ 1 if candidate $c \in C$ is placed as a shortcut

$f_e^q \in [0, 1]$ flow of demand q over edge $e \in E_{\text{ring}} \cup C$

$W \geq 0$ worst-case path length (the objective)

The optimisation graph consists of the fixed ring edges, always present with their true lengths, plus one edge per candidate, present only if selected, with weight L_c . All edges are traversable in both directions. The flow matrix is totally unimodular, as in the ring model, so the continuous relaxation is exact and no integrality needs to be imposed on f .

4 Phase 2 — Constraints

(D1) Node occupancy. At most one shortcut per node:

$$\sum_{c: v \in \{i(c), j(c)\}} z_c \leq 1 \quad \forall v \in V. \quad (1)$$

If the router returns several non-dominated variants for the same endpoint pair, they are kept as separate candidates; **(D1)** then automatically permits at most one of them, since both occupy the same two nodes.

(D2) Crossing limit. Each shortcut may cross at most one other selected shortcut. Crossings are permitted — they form a crossing signal exchanger and cost one additional wavelength — only exceeding the budget is forbidden:

$$\sum_{c' \in \mathcal{X}(c)} z_{c'} \leq 1 + |\mathcal{X}(c)| (1 - z_c) \quad \forall c \in C. \quad (2)$$

If $z_c = 1$ the right-hand side is 1 and at most one crossing partner may be selected; if $z_c = 0$ the constraint is vacuous. This is the same rule and the same algebraic form as the master’s **(M9)**; the difference is only the source of $\mathcal{X}$ — here it comes from real routed paths, in the master from the L-shape screen on always-crossing pairs.

(D3) Flow conservation. For every demand q and node v , over the combined ring-plus-candidate edge set:

$$\sum_{e \in \Delta^+(v)} f_e^q - \sum_{e \in \Delta^-(v)} f_e^q = \begin{cases} +1 & v = s(q) \\ -1 & v = t(q) \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

(D4) Coupling. A demand may use a candidate edge only if it is placed:

$$f_c^q \leq z_c \quad \forall q \in Q, \forall c \in C. \quad (4)$$

Ring edges need no coupling — they always exist. Note there is deliberately no per-shortcut capacity: any number of demands, in either direction, may share a placed shortcut. Shared use is the intended behaviour and is what lets a shortcut placed for one demand also benefit others.

(D5) Worst-case bounding, with w_e the true ring-edge length for $e \in E_{\text{ring}}$ and $w_c = L_c$ for candidates:

$$W \geq \sum_{e \in E_{\text{ring}} \cup C} w_e f_e^q \quad \forall q \in Q. \quad (5)$$

5 Objective

$$\min W \implies W_{\text{true}}(R). \quad (6)$$

No secondary summation term, no ε -regulariser, no wavelength term. Wavelength counting remains a postprocess outside the optimisation.

Caveat

A pure min–max objective admits large plateaus of equally optimal solutions, so the solver may return any member of that set. This is accepted. If a deterministic tie-break becomes necessary, prefer fewer bends on the worst-case demand’s path — but as a post-selection rule, not as a term in the objective.

6 Relationship to the proxy master

	Ring Proposer	Shortcut Solver
Ring	decision variables (b, r)	fixed input
Shortcut c	abstract endpoint pair	real routed path
Length charged	$\delta(c)$ (optimistic)	$L_c \geq \delta(c)$ (true)
Ring interaction	not modelled	obstacle during routing
Crossing data	from the L-shape screen	from actual routed paths
Result	LB , a lower bound	$W_{\text{true}}(R)$, a feasible value

Because $L_c \geq \delta(c)$ and the master omits geometric feasibility, the master’s optimum bounds the Shortcut Solver from below on every ring. It is the first stage in the pipeline that produces a genuinely realisable objective value.

7 Dual-safe early abort

The Shortcut Solver is expensive, so the outer loop may abort a Phase-2 solve early. Both of the following must hold, with ObjBound the solver’s own lower bound on $W_{\text{true}}(R)$:

$$\text{ObjBound} \geq W^* \quad \text{and} \quad \text{ObjBound} > LB_k. \quad (7)$$

Meaning: this ring can neither improve the incumbent nor produce a direct match. Both are required — if the solve cannot beat W^* , continuing may be pointless, but as long as its value could still fall to LB_k a direct match is still reachable and the solve must continue. Aborting therefore speeds up the outer loop without weakening its proof logic. An aborted run’s returned value must not be read as an exact fixed-ring optimum.

Notes

- **Cost profile.** Phase 1 dominates: the grid is built once, but every surviving candidate needs a grid shortest-path run, and the crossing matrix is $O(|C|^2)$ cheap intersection tests. Phase 2 is comparatively light — $|C|$ binaries plus a totally unimodular flow block, and no subtour elimination, since the ring is fixed.
- **Metric.** Plain Manhattan millimetres throughout, matching the master and the shared worst-case metric. Bend counts are recorded but do not enter W .
- **Bidirectionality.** A placed shortcut is usable in both directions by any demand. In the downstream wavelength model, however, the two directions are two independent waveguides and do not conflict with each other.